

University of Management and Technology

Course: Cloud Security

Title: Cloud Logging, Monitoring & SIEM

Group Number: 04

Instructor: Sir Ali Mansoor Elahi

Semester :6th

Submission date: 8 June 2026

Group Members

Student Name	Student ID	Assigned Role
Faiza Tauheed	F2023408075	Group Lead
Fawad Khan Lodhi	F2023105048	Literature Review Lead
Haris Waheed	F2023408001	Technical Lead
Hamza Zohaib	F2023408106	Documentation Lead
Hassaan Ahmed	F2023408017	Presentation Lead

1. INTRODUCTION

Cloud computing has changed the way organizations use and manage applications. These days cloud systems make an amount of data like logs and security information. This data helps us see what is going on in the system if there are any security problems and how well everything is working.

Cloud Logging is when we collect, store and look at the records of things that happen on resources, applications, networks and users. We need to keep an eye on everything all the time to make sure it is working performing well and is secure. There are systems called Security Information and Event Management that put all the logs together to find threats and help us respond to incidents.

More and more organizations are using clouds and mixing different types of clouds so the old ways of watching and managing are not good enough. The people in charge of security like those in Security Operations Centers use these systems to find guys look for threats and make the system respond automatically. We need to watch everything in time and find things that are not normal because bad people can attack really fast.

The Cloud Security Alliance made a list of security things and Logging and Monitoring is one of the main ones. If we log everything, we can be accountable follow rules look into problems find threats and respond to incidents.

It is really important to log and watch cloud things because organizations want to find security risks before they cause problems. They do this by watching cloud workloads what users are doing, network traffic and application events all the time to stay safe. Cloud logging and monitoring are key, to keeping everything secure.

2. PROBLEM STATEMENT

Organizations that use cloud services have a hard time gathering looking at and reacting to the number of security events that happen in their cloud. Some of the problems they face are:

Getting overwhelmed, with logs because they have many cloud services.

- Finding events is tough when there are millions of normal ones.
- It takes a time to detect threats because people have to monitor manually.
- They can't see what's happening across cloud services and environments.
- Traditional monitoring systems give many false alarms.
- Insider threats and advanced attacks can get past security controls.
- Attackers take advantage of monitoring, weak alerts, wrong logging settings and slow responses to get into cloud resources.

If organizations don't have SIEM and monitoring solutions they might experience:

Data breaches

- Financial losses
- Regulatory penalties
- Service outages
- Reputation damage

3. PROJECT OBJECTIVES

The main goals of this project are to do a thing.

First, we want to learn about the ways that cloud logging and cloud monitoring work.

We have following objectives for this project.

- We want to study cloud logging and cloud monitoring mechanisms.
- We need to look at what role SIEM platforms play in keeping the cloud secure.
- We have to see how real-time monitoring and getting alerts when something goes wrong actually work.
- We want to see which methods, for finding anomalies in the cloud are the best.
- We will compare the cloud-native SIEM solutions to see which one is the best.
- We will look at real life examples of cloud monitoring to see what works.
- We will take what we learned. Match it with CCM LOG controls and CAIQ concepts.
- Finally, we want to suggest ways to make cloud SOC operations better.

4. LITERATURE REVIEW

Cloud systems produce a lot of data that's hard to handle with traditional methods. To deal with these researchers are working on using intelligence and machine learning in cloud systems.

4.1 Machine Learning and Deep Learning Foundations

New developments are using models that do not need a lot of labeled data to understand log behaviors. For example, Sefati and his team made a framework called SSLA that can analyze web application logs in time. This framework does not need a lot of labeled data. Jilcha and his team used a kind of loss function with Transformer networks to detect anomalies without any prior knowledge. They used this to handle changes in infrastructure. Some people are also using architectures that combine LSTMs and Transformers for cyber forensic investigations. However these architectures use a lot of computing power. Yang and Harris used a LLaMA-based architecture to track logs and understand their meaning. In settings Ogunmolu and his team used a deep autoencoder to detect threats in Kubernetes audit logs. This increased the recall of threat detection from 54% to 84%.

4.2 Integrated Zero Trust and Security Automation Architectures

The literature shows that we need to be proactive in our approach. A good framework is ZenGuard, which combines SIEM, SOAR and UEBA into an open-source Zero Trust architecture. ZenGuard uses Isolation Forests and One-Class SVMs to reduce the time it takes to respond to attacks. For cluster execution security ShadowKube introduced runtime behavioral monitoring paired with shadow honeypots. This helped to prevent 43 attacks in production Kubernetes clusters. These platforms help to translate telemetry insights into automated cloud-native remediation policies.

4.3 Multi-Cloud Infrastructure and AI Observability

As companies use cloud systems it is important to normalize telemetry data across different layers. Johnson and Antwi made frameworks to secure -cloud environments while following regulations, like GDPR and HIPAA. Finally as large language models are used in cloud systems we need platforms that can observe and monitor them. Sisodia introduced safety guardrails to detect injections while the LumiMAS framework used localized autoencoders to detect threats quickly without incurring high costs. Systems and cloud architectures are using machine learning and artificial intelligence to improve security and detection. Cloud architectures are becoming more important in the use of cloud systems.

5. TECHNICAL BACKGROUND

5.1 Cloud Logging Fundamentals

Cloud logging is the base for all monitoring and security information and event management or SIEM. In environments logs come from almost every service and action. This includes things like API calls, user authentication, network flow data and application errors. Knowing the types of cloud logs is crucial, for building good security monitoring systems.

5.1.1 Types of Cloud Logs

Log Type	Description	Availability	Cloud Provider Tool	Security Use Case
Admin Activity Logs	Records API calls modifying resource config	All cloud providers (always enabled)	CloudTrail (AWS); Activity Log (Azure); Cloud Audit Logs (GCP)	Unauthorized config changes, privilege escalation
Data Access Logs	Logs read/write access to sensitive data	Optional; must be enabled manually	S3 access logs, Azure Blob logs, GCS data access logs	Data exfiltration, unauthorized reads
System Event Logs	OS-level events; process creation, logins	VM/instance level logging agents	CloudWatch Logs Agent, Azure Monitor, Stackdriver	Malware, lateral movement, insider threats
Network Flow Logs	Captures IP traffic metadata (src, dst, port)	VPC/VNet level; enabled per flow log policy	VPC Flow Logs (AWS), NSG Flow Logs (Azure), VPC Flow (GCP)	Port scans, DDoS patterns, C2 traffic

Application Logs	App-level events (errors, auth, transactions)	App framework or custom instrumentation	CloudWatch App Logs, Log Analytics Workspace	Business logic attacks, injection attempts
------------------	---	---	--	--

5.1.2 Log Management Architecture

Cloud log management is like a process that we need to follow. It has a steps: we need to collect all the raw log data from every source then we have to put all these logs in one place, which is called centralization. After that we have to make sure all the logs are in the format this is what we call normalization. We also need to add information to the logs like where the threat is coming from where it is located and what asset it is trying to attack this is called enrichment.

Then we have to store all these logs for a while so we can look at them later. Make sure we are doing everything we are supposed to be doing. We usually keep them for 90 days where we can look at them easily and for one year where they're not as easy to look at. After that we need to look at all the logs and see if there is anything that does not look right this is what we call analysis. And if we find something that does not look right we need to let people know this is called alerting. It happens when the cloud log management system finds a pattern that could be a security threat.

5.2 Security Information and Event Management (SIEM)

Security Information Management, which is focused on storing logs for a time and making reports and Security Event Management, which is focused on looking at events in real time and sending out alerts. The new SIEM platforms do all these things. Also include advanced analytics, threat intelligence integration and automated response orchestration. This means the Security Information and Event Management system can look at all the logs and events find patterns and automatically respond to security threats. The Security Information and Event Management system is very important for cloud log management and, for keeping our systems secure.

5.2.1 Core SIEM Components

- **Log Collection and Aggregation Engine:** We get log data from sources like syslog, API connectors, agents and cloud services such as AWS Kinesis, Azure Event Hub and GCP Pub/Sub. These logs can be in formats like JSON, CSV or even plain text.
- **Data. Parsing:** This step converts all log formats into one standard format. We use schemas like Common Event Format (CEF) Log Event Extended Format (LEEF) and Elastic Common Schema (ECS). This helps us to easily compare logs from sources.
- **Correlation Engine:** Here we use rules to find connections between events from log sources over time. For example, if there are 10 failed logins in 5 minutes. If a user logs in and then tries to escalate privileges. We also look for behavior that deviates from a user's normal activity.
- **Threat Intelligence Integration:** We add threat intelligence feeds to our events. These feeds come from sources like MISP, Virus Total and other commercial threat intelligence providers. They help us detect known threats and indicators of compromise (IOCs).
- **Alert Management and Case Management:** When we find something we generate a security alert. We prioritize these alerts. Help analysts investigate and escalate incidents. We also integrate with ticketing systems like ServiceNow and Jira.
- **Compliance and Reporting:** We have pre-built dashboards and reports for regulatory frameworks like PCI-DSS, HIPAA, GDPR, SOX, ISO 27001 and NIST. This helps us evidence collection for audits and ensures we are compliant, with these regulations.

5.2.2 SIEM Deployment Models

- On-Premises SIEM is a type of security system that we deploy inside our company's data center. This gives us control over our data and we can do what we want with it. It costs a lot of money to set up and we need special people to take care of it. For example, we have IBM QRadar that we use inside our company and Splunk Enterprise.
- Cloud-Native SIEM is a kind of security system that we get from a cloud service. This means it can grow as we need it to. It is easy to set up. It also works well with the cloud providers systems. For example, we have Microsoft Sentinel that works with Azure and AWS Security Lake with Amazon OpenSearch and Google Chronicle Security Operations.
- Hybrid SIEM is a mix of the two systems I just talked about. We collect log information from inside our company. Then we send it to the cloud to be looked at and stored. This is good for companies that have some things inside their buildings and some things in the cloud. For example, we have Splunk Cloud that works with helpers inside our company.
- Open-Source SIEM is a type of security system that people work on together and share with everyone. This means we do not have to pay to use it. For example, we have Wazuh and Security Onion and OSSIM from AlienVault and ELK Stack which includes Elasticsearch and Logstash and Kibana.

5.3 Key Technical Concepts

5.3.1 Security Orchestration, Automation and Response (SOAR)

Security teams use SOAR to make their SIEM tools better by adding automated playbooks to their security workflows. When the SIEM tool finds a problem the SOAR platform can do things on its own like isolate bad instances take away bad user credentials block bad IP addresses get memory dumps and tell people who need to know. All without a person having to do it. This makes it much faster to respond to problems, which's very important in cloud environments where bad things can happen very quickly.

5.3.2 User and Entity Behavior Analytics (UEBA)

UEBA uses machine learning to learn what is behavior for users, devices and service accounts in organizations cloud environment. Then it checks to see if anything is doing something that's not normal. If it finds something that's not normal it can tell us if someone's account has been

compromised, if someone inside the organization is doing something bad or if someone is using their power in a bad way. Even if we do not know what the bad thing looks like. UEBA is very good at finding sneaky attacks that the SIEM tool might miss.

5.3.3 Log-Based Anomaly Detection with Machine Learning

The old way of using SIEM tools relies on people making rules that the tool has to follow. This is hard to do and it is hard to make new rules when new bad things happen. Machine learning can help with this by looking at log data and finding things that are not normal. There are a way to do this such as using special models like Deep Log and LogBERT that look at log data over time and find things that are not normal. We can also use special language models like LogFiT and LogLLM that understand what log messages mean and find things that do not make sense. Another way is to use methods like k-means and isolation forest that find log events that are not, like the others. We can use graph-based methods that look at how log events are related to each other and find things that are not normal.

5.3.4: Cloud-Native Monitoring Tools and Services

AWS CloudTrail: Captures all calls made to all AWS services, giving an audit trail of user, role and service actions. Essential for governance, compliance, and security investigation in AWS environments.

AWS GuardDuty: Threat detection service with managed ML models that constantly watch CloudTrail, VPC Flow Logs and DNS logs for detection of threats such as cryptocurrency mining, data exfiltration, and account compromise.

Microsoft Sentinel: Cloud-native SIEM and SOAR, powered by Azure, with AI-driven threat detection, automated investigation workflows and the ability to connect with the entire Microsoft security ecosystem.

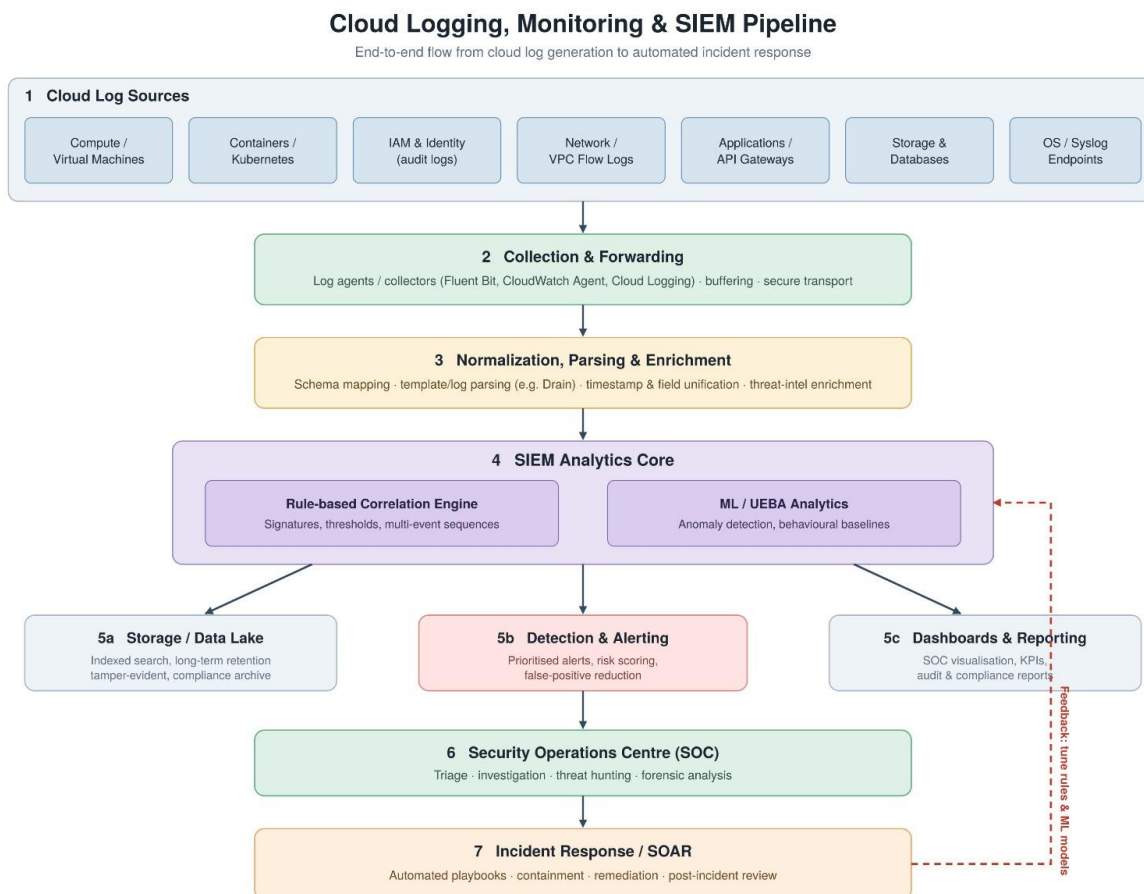
Azure Monitor and Defender for Cloud: Offers comprehensive observability and posture management of your Azure workloads, which includes compliance assessment, vulnerability scanning, and threat protection.

Google Chronicle Security Operations: A cloud-based security analytics platform designed using Google's data technology that allows petabyte-scale log storage and sub-second search on years of security data.

Google's centralized security and risk management platform for Google's Cloud Platform (GCP) environments, including asset inventory, vulnerability assessment, threat detection, and compliance monitoring.

6. ARCHITECTURE / WORKFLOW DIAGRAM

Cloud SIEM Monitoring Workflow



7. PRACTICAL IMPLEMENTATION / CASE STUDY

Case Study: Microsoft Sentinel

Microsoft Sentinel helps organizations keep an eye on their security by collecting information from sources.

Working Process

1. It gathers logs from places like Azure, AWS and Microsoft 365 well as, from endpoints and applications.
2. Then it makes all the incoming events look the same.
3. Next it applies rules to check for activity.
4. If something fishy is found it flags it as a threat.
5. This potential threat is turned into an incident that needs attention.
6. It also supports automated workflows to respond to these incidents.

Practical Scenario

Attack Simulation includes:

- Multiple failed attempts to log in
- Logins from locations
- Events where someone tries to gain more privileges

Detection Process:

Microsoft Sentinel looks at all the events together.

- It uses rules to spot patterns of behavior like repeated failed logins.
- When it finds something it sends out an alert.
- A security analyst then investigates the alert.
- If needed a response workflow is triggered to block the activity.

Observations

- It provides visibility into potential threats.
- It reduces the time it takes to respond to security incidents.
- It allows for monitoring of security events.
- It improves reporting for compliance purposes.

COMPARATIVE ANALYSIS

Paper / Framework	Core Technique	Key Finding / Metric	Major Limitation	Environment
SSLA (2025)	Semi-Supervised Learning	High real-time injection detection with minimal data labeling.	Limited scope; web app logs only.	Web Applications.
Temporal Decay (2025)	Transformers + Temporal Loss	Robust zero-shot anomaly detection capability.	Highly computationally expensive.	Enterprise SOC's.
Advanced Log Analyzer (2025)	LSTM + Transformer	Excellent sequential tracking for post-incident forensics.	High overhead; demands pristine logs.	Cyber Forensics.
Log Transformers (2025)	LLaMA LLM + RL	Superior linguistic and semantic log understanding.	Expensive training and hosting costs.	Next-Gen AI SIEM.
ZenGuard (2025)	SIEM + SOAR + UEBA via ML	Collapsed MTTR to under 10 seconds during attacks.	Risk of baseline drift and processing latency.	Zero Trust Cloud.
ShadowKube (2025)	Behavioral Tracking + Honeypots	Detected 635 attacks across 43 severe CVEs.	No serverless support; high compute strain.	Kubernetes (K8s).
LumiMAS (2026)	Localized Deep Autoencoders	Under 0.06s anomaly oversight with zero external API fees.	Massive node-level CPU/memory draw.	Multi-Agent AI.
AISOC Fusion (2026)	Cross-Layer Data Fusion	Dramatically reduces false alarms on multi-layer attacks.	Puts severe resource strain on microservice nodes.	Cloud Microservices.
AI Observability (2026)	Unsupervised Autoencoder	Elevated threat detection recall from 54% to 84%.	Entirely dependent on pristine telemetry pipelines.	Container Stacks.

9. Limitations

The system faces some challenges.

- It has to process a lot of logs at the time.
- We need to be able to see what is going on across cloud services.
- We have to make sure all the data is, in a format that we can use.
- It is hard to get all the different parts of the system to work together.

The security of the system has some limitations.

- We get many alerts and it is hard to know which ones are important.
- Sometimes the system tells us about a problem when there is not one.
- People who work for the company can sometimes cause security issues.
- The system has trouble detecting attacks that use encryption.

The cost of the system is also a challenge.

- It costs a lot of money to use a security information and event management system.
- We have to pay to store all the data.
- It is expensive to keep the data for a time.

The system also has some scalability issues.

- The number of logs we have to process is getting bigger.
- We have to be able to handle a lot of data coming in quickly.

There are also some human-factor risks.

- We do not have people to analyze the data.
- The people we have do not always have the skills they need.
- Sometimes the system is not set up correctly.

10. FUTURE WORK

- We think that the system will get better in the future.
- Future developments are expected in security operations centers that use intelligence.
- We will be able to investigate and respond to security issues automatically.
- We will use language models to do security analytics.
- We will be able to use language to hunt for threats and explain alerts.
- We will be able to detect threats before they happen.
- We will have security operations that can run on their own.
- We will be able to create environments that can fix themselves.
- We will be able to do behavioral analytics.
- We will be able to analyze what users and devices are doing.

We will be able to see what is going on across cloud services, including Amazon Web Services, Azure and Google Cloud.

11. CONCLUSION

Cloud logging and monitoring and SIEM systems are the basics of how we keep our cloud security running. As our cloud environments get more complicated, we need ways to monitor them so we can see what is going on and find any threats.

This study looked at how cloud logging's set up what SIEM technologies are out there how SOC operations work how we can monitor things in real time what alerting mechanisms we can use and how we can detect anomalies. What we found out from looking at the literature is that people are moving towards using intelligence to detect threats and automate our security operations.

The CCM Logging and Monitoring domain is about making sure we collect events securely monitor them and audit them. Organizations that use cloud-native SIEM platforms, automated alerting and monitoring that is assisted by artificial intelligence can really improve how secure they are and how they respond to incidents.

We recommend that organizations use logging implement cloud-native SIEM solutions keep monitoring all the time use artificial intelligence to detect anomalies and regularly check their controls using the CCM and CAIQ frameworks. Cloud logging and monitoring and SIEM systems are very important, to our security operations.

12. REFERENCES

- Cloud Security Alliance. (2025). Cloud Controls Matrix (CCM) v4.0: Logging and Monitoring (LOG) Domain Enforcement Guidelines. Cloud Security Alliance Assessment Repository. Antwi, N. W. (2025). Threat Detection in MultiCloud Environments. *Advanced Cloud Security Architectures and Compliance Governance*, 112–135.
- Arif-*, T., Jo, B., & Park J. H. (2026). A Comprehensive Survey of Privacy-Enhancing and Trust-Centric Cloud-Native Security Techniques Against Cyber Threats. *IEEE Xplore / MDPI Sensors*, 26(4) 1042–1065. Arul Selvam, P. (2025). Explainable AI for SIEM Alert Correlation and Threat Hunting. *Computation: Experience*, 37(12) e70390. <https://doi.org/10.1002/cpe.70390>.
- Borges, M. C., & Werner, S. (2026). Continuous Observability Assurance in Cloud-Native Applications. *Proceedings of the ACM/IEEE International Conference on Cloud Engineering (ICCE)* arXiv:2603.08941.
- Hassan, A., Rauf, A., Shafqat, N., Latif, R., & Khan, H. (2025). ZenGuard: A Machine Learning Based Zero Trust Framework for Context-Aware Threat Mitigation Using SIEM, SOAR and UEBA. *Journal of Network and Computer Applications*, 242 103980.
- Jilcha, L. A., et al. (2025). Temporal Decay Loss for Adaptive Log Anomaly Detection in Cloud Environments. *Sensors*, 25(9) 2649. <https://www.mdpi.com/1424-8220/25/9/2649>.
- Johnson, E., et al. (2025). Deep Learning-Based Security Monitoring in Multi-Cloud Environments. *ResearchGate Technical Reports*, Publication ID: 394471956.
- Lee, J., et al. (2025). AI-Driven Log Analytics for Cloud Security Operations Centers. *Applied Sciences*, 15(13) 7237. <https://doi.org/10.3390/app15137237>.
- Nipa, N. A., Bouguila, N., & Patterson Z. (2025). A Comparative Study of Log-Based Anomaly Detection Methods in Real-World System Logs. *IEEE Transactions on Reliability*, 74(2) 589–604.
- Ogunmolu, A. M., Aroh, I. S., Henry, O., Adeyinka, P. D., & Olutimehin A. T. (2026). AI-Driven Observability for Managing Security Complexity in Cloud-Native Microservices and Containerized Environments. *International Journal of Cloud Computing & Observability* 5(1) 45–68

- Okonkwo, N. P., & Dhirani L. L. (2025). Cloud-Native SIEM Architectures for Threat Detection and Incident Response. ArXiv Preprints, arXiv:2512.14935.
- Okonkwo, N. P., & Dhirani L. L. (2026). Cloud Security Leveraging AI: A Fusion-Based AISOC for Malware and Log Behavior Detection. International Journal of Information Security, arXiv:2601.12490.
- Rossi, A. G. (2025). Cloud-Native. Observability Frameworks for Modern Digital Transformation Initiatives. Journal of Systems and Software Architecture, 164 103412.
- Sefati, S. S., et al. (2025). SSLA: A Semi-Supervised Framework for Real-Time Injection Detection and Anomaly Monitoring in Cloud-Based Web Applications. Journal of Cloud Computing, 14(1) 32. <https://link.springer.com/article/10.1186/s13677-025-00765-6>.
- Sisodia, T. (2025). AI Observability for Large Language Model Systems. IEEE Open Journal of Computer Science, arXiv:2508.11304.
- Solomon, R., Levi, Y. Y., Vaknin, L., Aizikovich, E., Baras, A., Ohana, E., Giloni, A., Bose, S., Picardi, C., Elovici, Y., & Shabtai A. (2026). LumiMAS: A Comprehensive Framework for Real-Time Monitoring and Enhanced Observability in Multi-Agent Systems. ArXiv Preprints / CS.CR arXiv:2602.04312.
- Yang, Z., & Harris I. G. (2025). Log-Based Anomaly Detection Using Transformer Architectures in Cloud Systems. ArXiv Preprints, arXiv:2503.14849.
- Anonymous. (2025). Advanced System Log Analyzer for Anomaly Detection and Cyber Forensic Investigations Using LSTM and Transformer Networks. Journal of Cloud Computing, 14(1) 54. <https://link.springer.com/article/10.1186/s13677-025-00789-y>.
- Anonymous. (2026). FSECAD: Feature-Selected Explainable Cloud Anomaly Detection Framework. Scientific Temper, 17(1) 89–101.
- Anonymous. (2026). Integrated Methodology for Intrusion Detection and Mitigation to Optimize Cloud Security. Journal of Cloud Computing, 15(1) 112. <https://link.springer.com/article/10.1186/s13677-026-00858-w>.
- Anonymous. (2025). On Translating Monitoring Insights into a Cloud-Native Execution Environment. Proceedings of the International Conference on Cloud Computing, IEEE Xplore.
- Anonymous. (2025). ShadowKube: Enhancing Kubernetes Security with Behavioral Monitoring and Honeypot Integration. Proceedings of the USENIX Security Symposium, 304–321.
- Anonymous. (2025). Cloud Security Automation Through Symmetry: Threat Detection and Response. Journal of Cybersecurity and Automation 8(3) 145–162.